P. R. Pote Patil College of Engineering and Management, Amravati

Title of Project

" Delivery Route Optimization using A* and Uniform Cost Search "

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B.Tech. Second Year -4th Sem

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CERTIFICATE

This is to certify that following group of students of have successfully completed the AI&DS Lab project titled **"Timeline of AI History using Python Libraries"** for the requirements of the Bachelors of Technology (B.Tech) program in AI&DS Laboratory during the academic year 2025–26.

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P. R. Pote Patil College of Engineering & Management, Amravati Session 2025 – 26

Abstract

The *Delivery Route Optimization using A* and Uniform Cost Search* project is designed to find the most efficient path between different locations in a city using intelligent search algorithms. The system focuses on minimizing travel distance and time by computing optimal routes on a graph-based map, where nodes represent locations and edges represent roads with associated costs such as distance or travel time.

The application enhances decision-making by applying two widely used pathfinding algorithms—A* and Uniform Cost Search (UCS)—and comparing their performance. While UCS guarantees the shortest path based on cost, A* improves efficiency by using a heuristic approach, reducing the number of nodes explored and improving execution speed.

The system provides an interactive and visual representation of routes, allowing users to input source and destination points and observe the computed optimal path along with distance and estimated time. It also demonstrates real-time processing of route data and highlights differences in algorithm performance, making complex concepts easier to understand.

The project is implemented using appropriate programming tools and graph-based data structures, ensuring efficient computation and scalability. It is designed to be simple, user-friendly, and practical, making it useful for students, beginners, and real-world applications such as delivery services and navigation systems.
By combining algorithmic efficiency with visualization, this project provides an effective solution for route optimization, improving delivery planning, reducing travel costs, and enhancing overall system performance.

ACKNOWLEDGEMENT

It is our sincere duty to express our gratitude to all those who have guided us during the preparation of our project.

First of all, we extend our heartfelt thanks to our respected Principal, **Dr. P. M. Jawandhiya**, for his support and encouragement throughout the project.

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Chapter 1

INTRODUCTION

1. Objective

i. To develop a system for finding optimal delivery routes using search algorithms.

ii. To implement **A*** and **Uniform Cost Search (UCS)** for path optimization.

iii. To compare the efficiency of both algorithms in terms of cost and performance.

iv. To represent locations and paths using graph-based data structures.

v. To calculate the shortest path based on distance or travel cost.

vi. To provide a simple and interactive interface for user input.

vii. To improve decision-making in route planning.

viii. To design a user-friendly and efficient system.

ix. To demonstrate the practical application of AI algorithms in real-world problems.

1. Introduction

Delivery Route Optimization is an important application of Artificial Intelligence that focuses on finding the most efficient path between different locations. This project uses intelligent search algorithms to determine the shortest and most cost-effective route for delivery services.

The system is based on graph representation, where locations are treated as nodes and paths between them are represented as edges with associated costs such as distance or time. Two important algorithms—A* and Uniform Cost Search (UCS)—are used to compute optimal routes. While UCS guarantees the shortest path, A* improves performance by using heuristic functions to guide the search process.

The project allows users to input source and destination locations and calculates the best possible route. It also helps in visualizing how different algorithms work and compares their efficiency.

This system is developed using suitable programming tools and provides a simple, interactive, and user-friendly interface. It helps in understanding complex concepts of pathfinding and optimization in an easy and practical way.

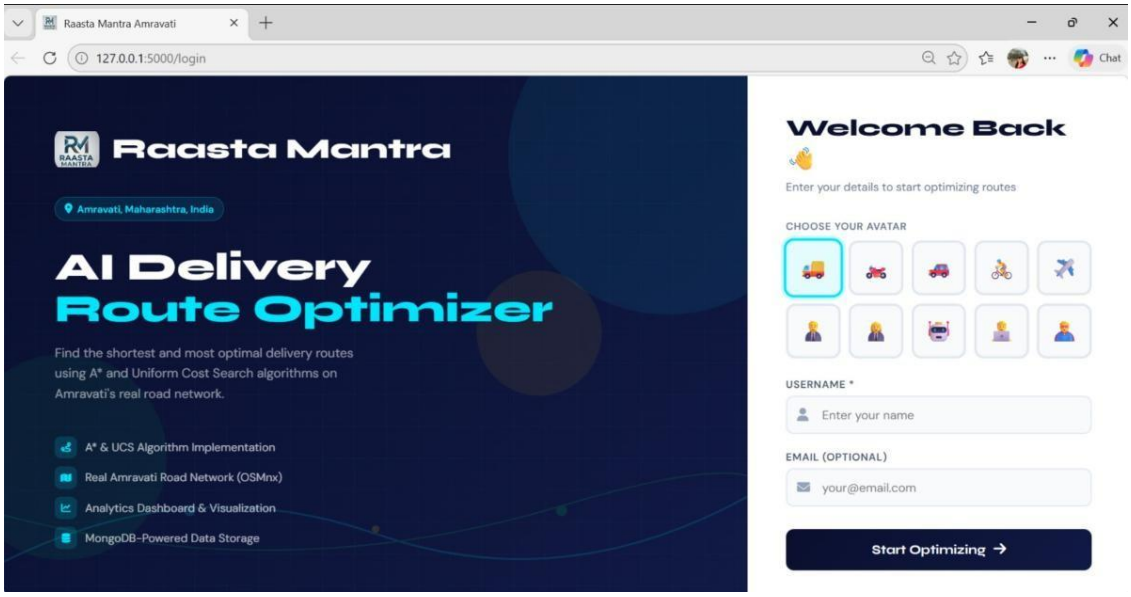

Figure 1.2.1: Sign In Page (Objective and Purpose)

Chapter 2

PROPOSED WORK

1. Basic Idea

The project aims to develop a system called *Delivery Route Optimization using A* and Uniform Cost Search*. The system is designed to find the most efficient route between different locations in a simple and interactive way. Instead of manual route planning, this application uses intelligent algorithms to automatically calculate the shortest and most cost-effective path.
In this system, the user selects a source and destination, and the system represents the map as a graph where locations are nodes and roads are edges with costs. The algorithms A* and Uniform Cost Search are applied to compute the optimal route. The system then displays the best path along with distance and route details.

Key features :

- The system is designed to make route planning easy, fast, and efficient using algorithm-based optimization.
- **User Input System**
  Users can select source and destination locations easily.
- **Graph-Based Route Representation**
  Locations and paths are represented using nodes and edges.
- **A* and UCS Algorithm Implementation**
  Both algorithms are used to find the optimal path.
- **Optimized Route Display**
  The shortest and best route is shown to the user.
- **Map Visualization**
  Routes are displayed visually on a map for better understanding.
- **Dashboard System**
  Displays route details and comparison of results.
- **Real-time Processing**
  Routes are calculated instantly based on user input.
- **Feedback System**
  Users can provide feedback about the system.
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- **Simple and User-Friendly Interface**
  Easy to use for beginners and students.

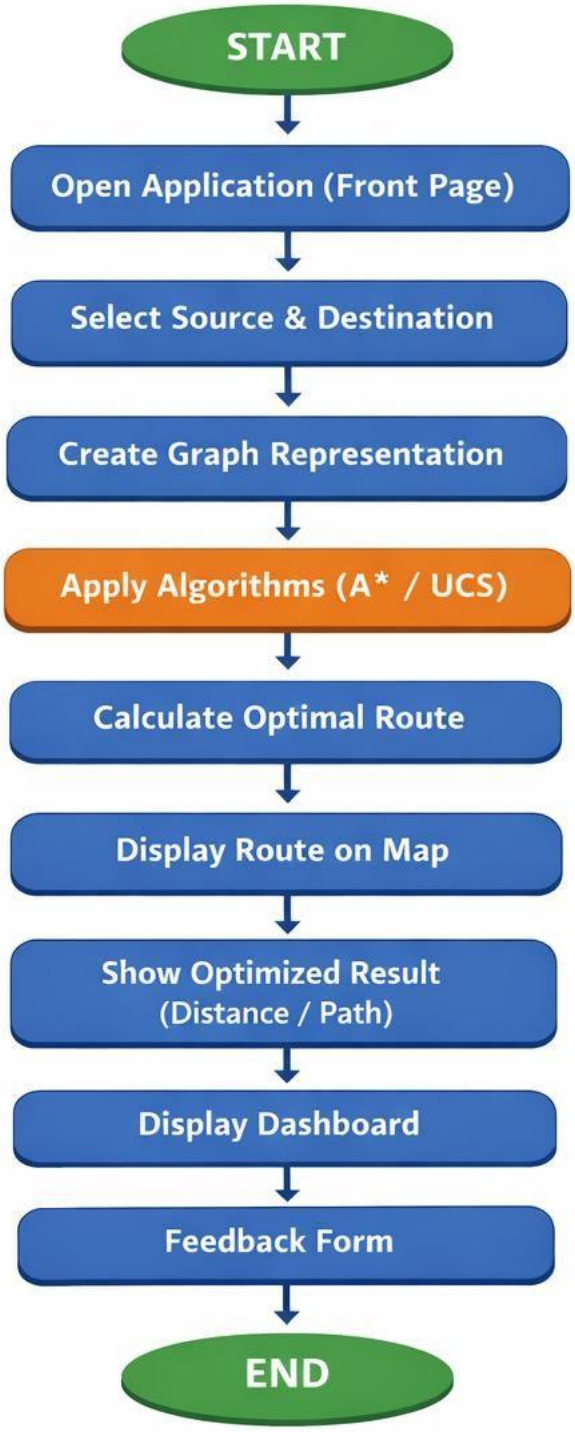


Figure 2.2.1: Workflow of Delivery Route Optimization System

2.3 Methodology

The methodology for developing the Delivery Route Optimization using A* and Uniform Cost Search follows a structured approach to design an efficient and user-friendly system for optimal route planning.

- **Development**

1. Frontend:

- HTML, CSS, JavaScript – Used to design a responsive and interactive web-based user interface for input and visualization.

2. Backend:

- Python – Used for implementing core logic and algorithm execution.
- Flask – Used to handle server-side processing and communication between frontend and backend.

3. Algorithms:

- A* Algorithm – Used to find the shortest path using heuristic-based search.
- Uniform Cost Search (UCS) – Used to compute the optimal path based on path cost.

4. Libraries:

- NetworkX – Used for graph creation and pathfinding operations.
- Matplotlib – Used for visualization of graph and routes.

5. Database:

- MongoDB – Used for storing and managing route data, user user inputs, and system information.

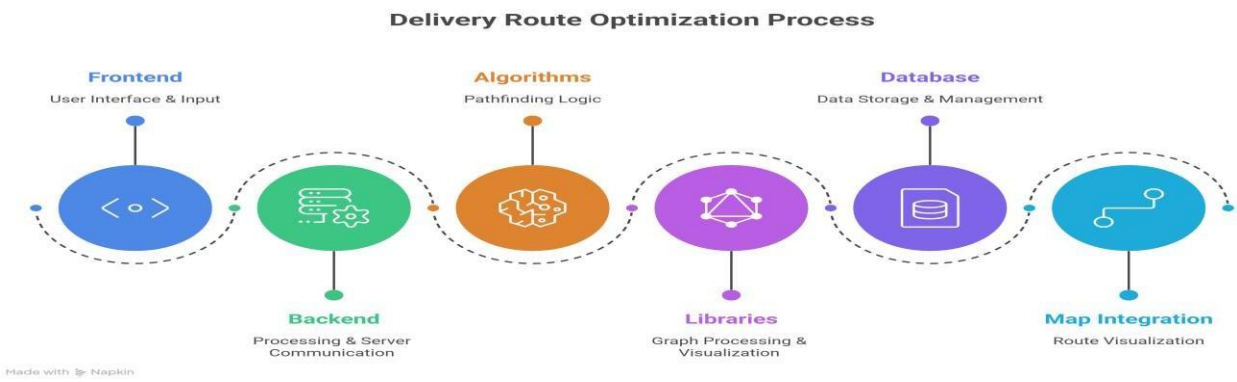


Figure 2.3.1: Methodology

Chapter 3

RESULT

1. Result

The Delivery Route Optimization using A* and Uniform Cost Search system was successfully developed and tested. The system works properly and helps in finding the shortest route between different locations.

Users can select source and destination, and the system shows the best route on the map along with distance. Both A* and UCS algorithms give correct results, but A* works faster in most cases.

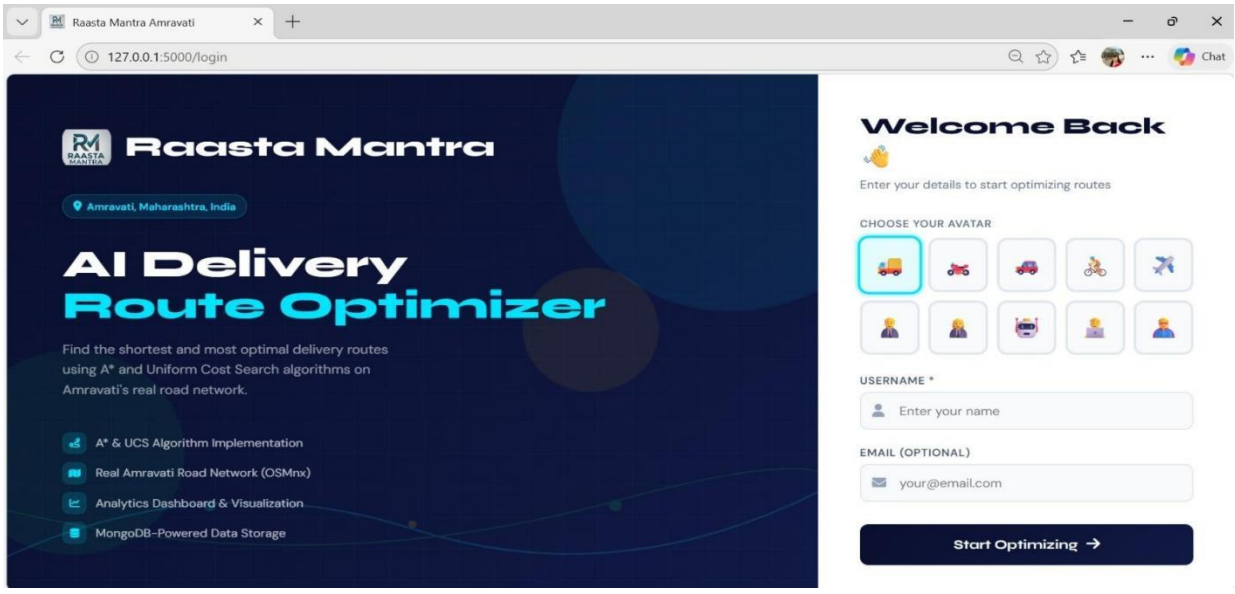


Figure 3.2.1 : Login/Register

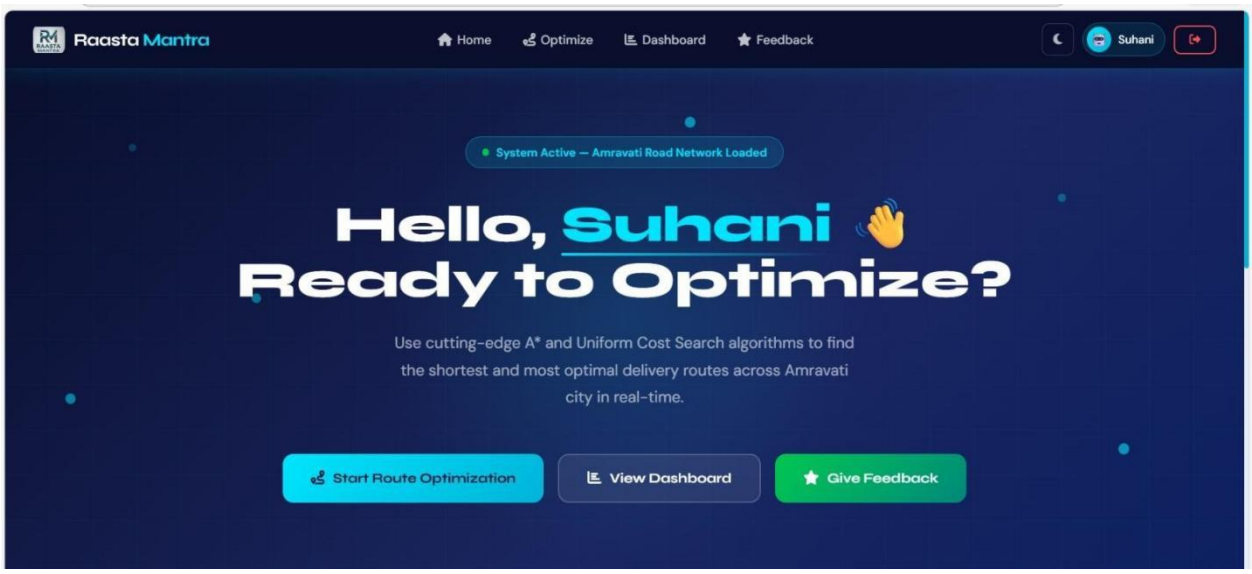


Figure 3.2.2 : User Dasboard

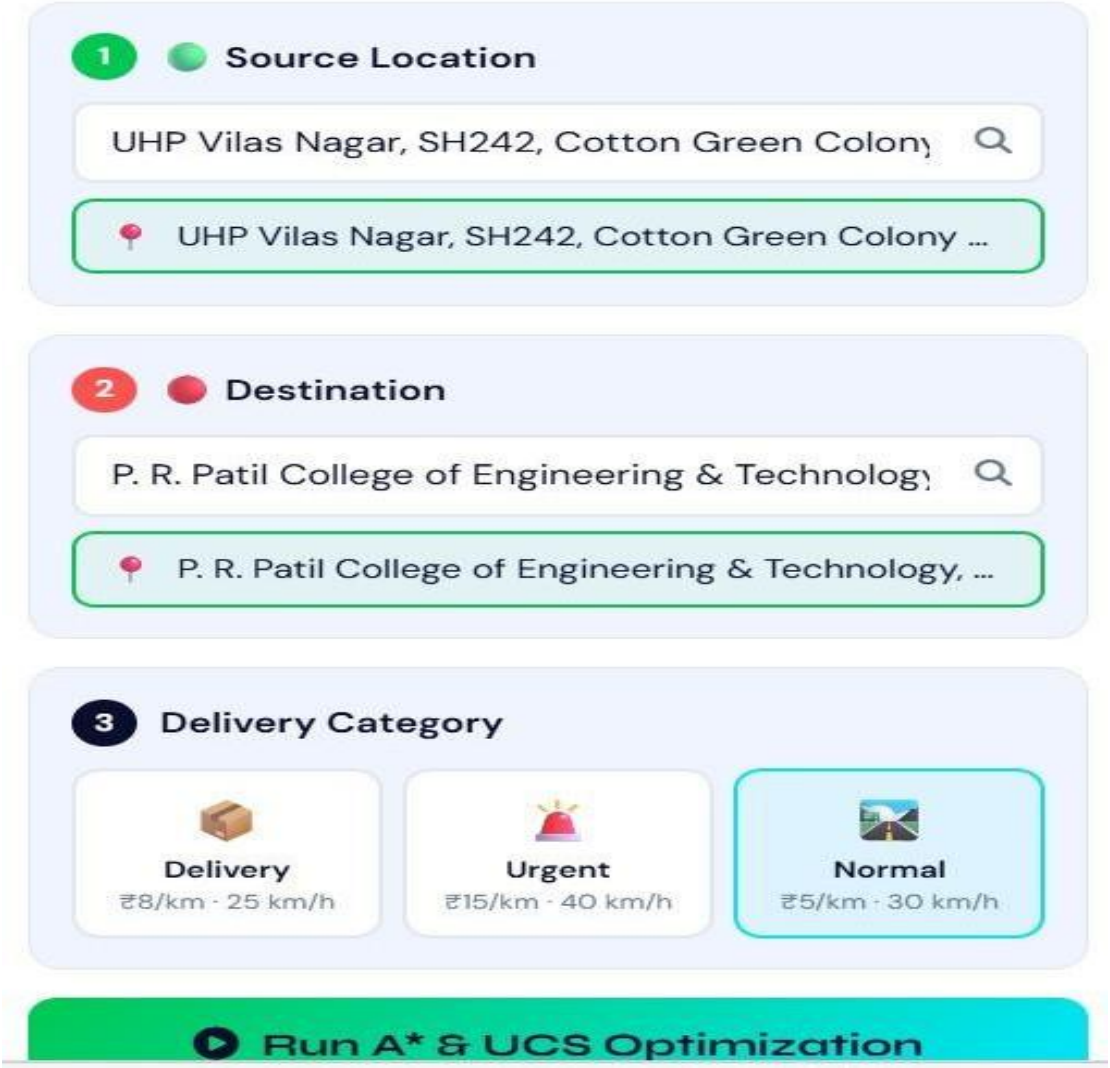


Figure 3.2.3: Route Planner

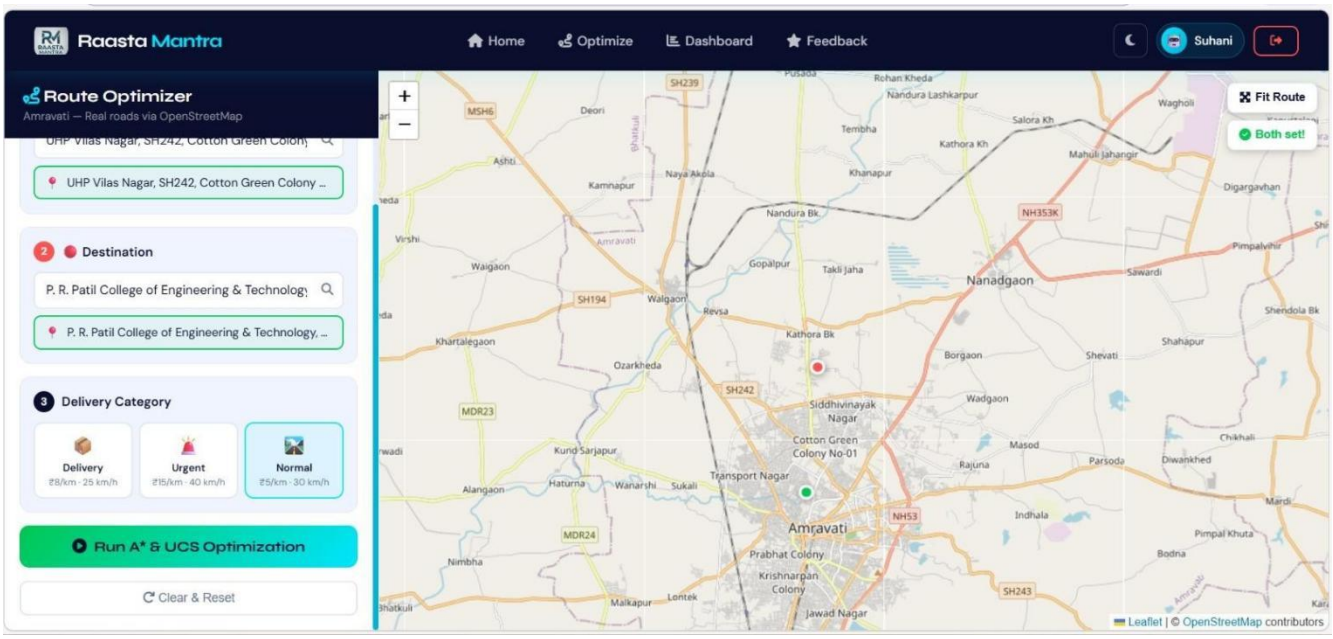


Figure 3.2.4: Map Interface

Figure 3.2.5: Pick Route

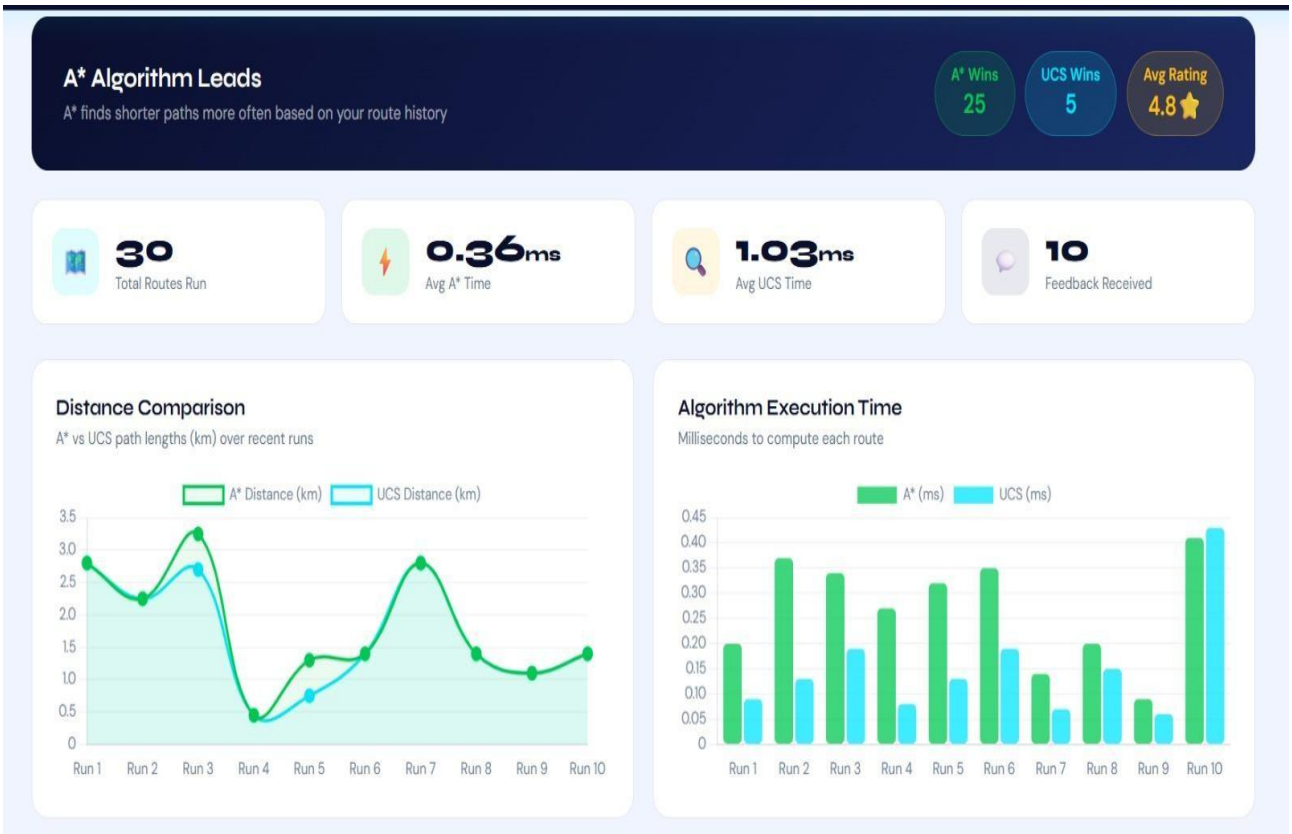


Figure 3.2.6: Performance Dashboard

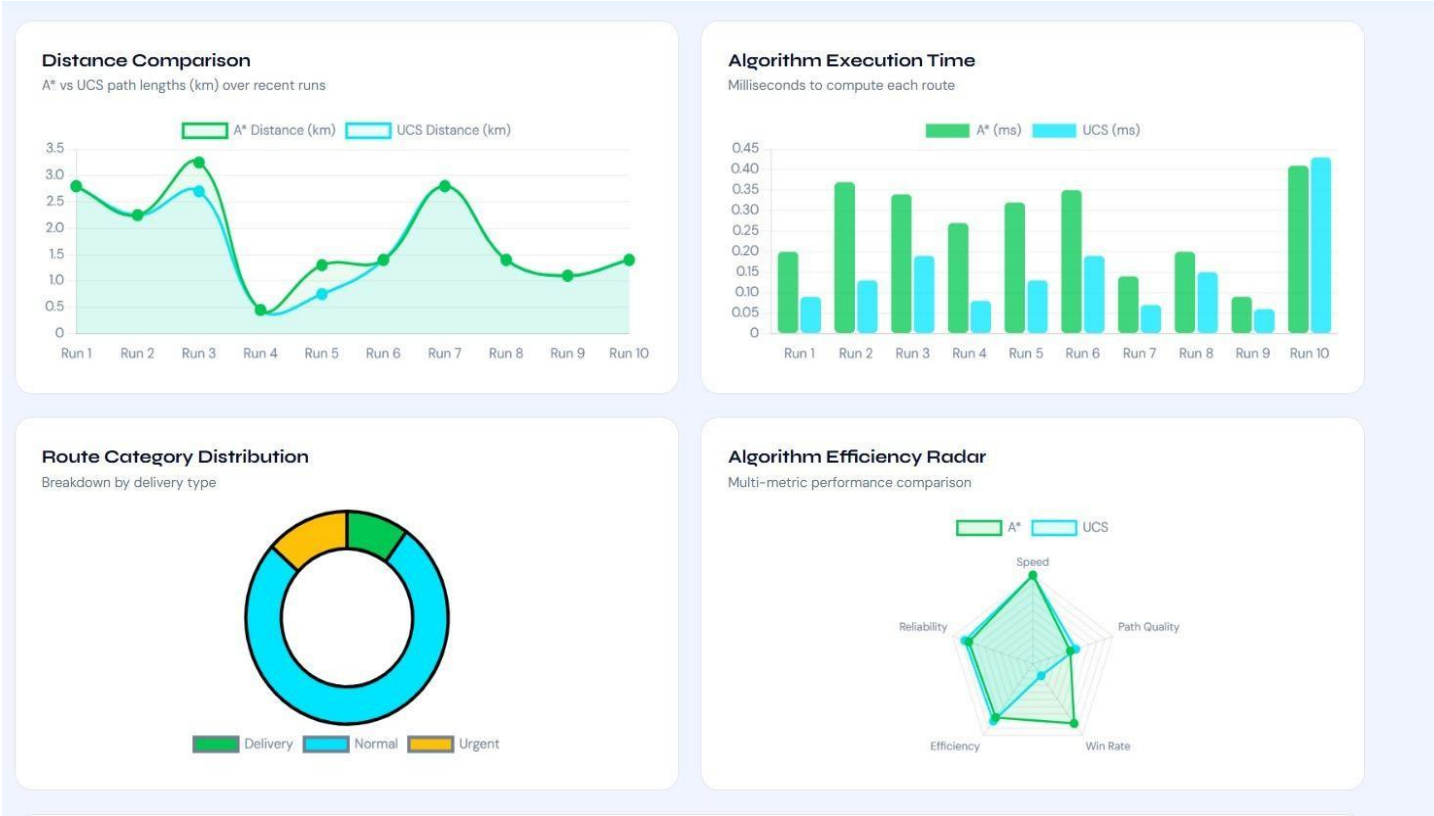
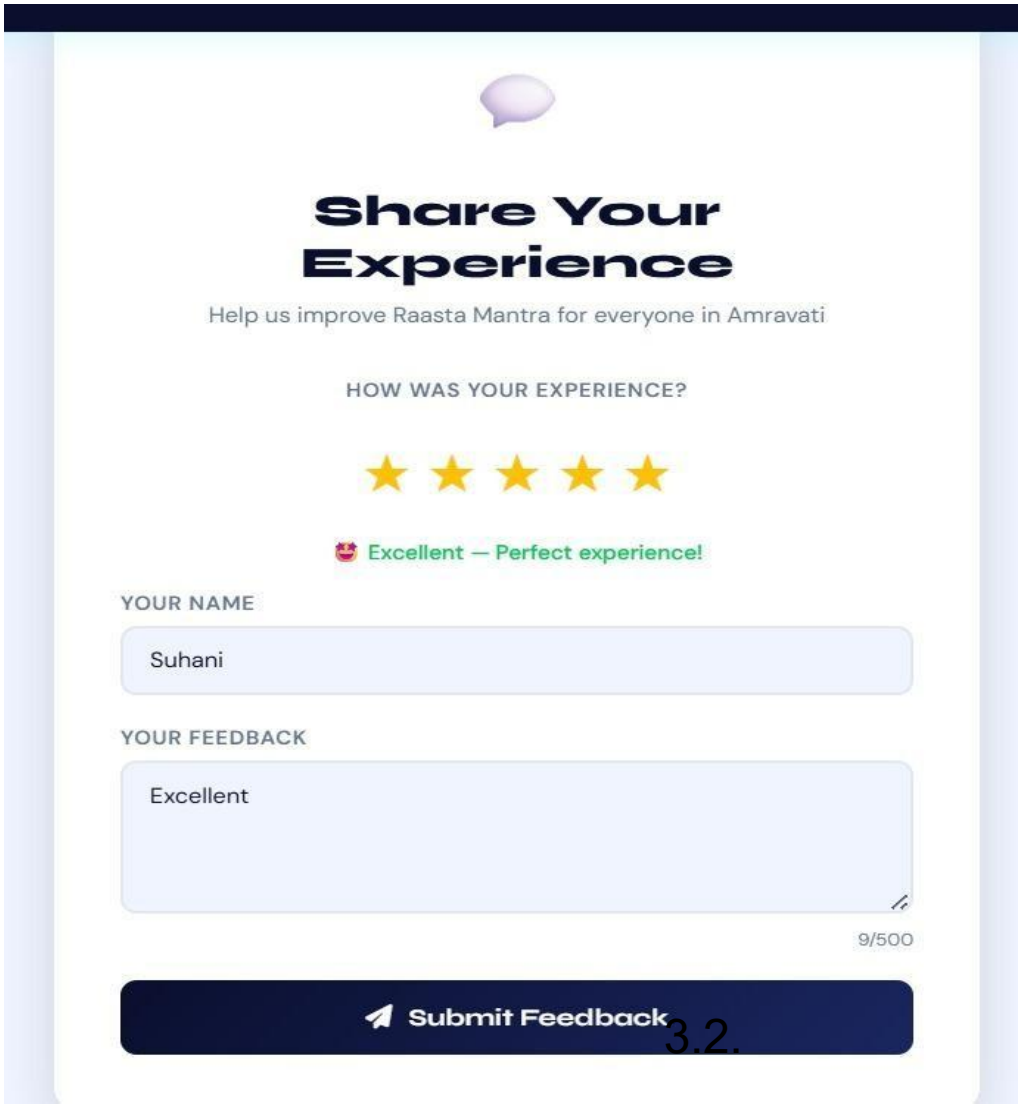


Figure 3.2.7 : Insights.



3.2.

Figure 3.2.8 : Feedback Form

Chapter 4
CONCLUSION

1. The Delivery Route Optimization using A* and Uniform Cost Search system was successfully developed. It provides a simple and effective way to find the best route between different locations.

2. The system uses A* and UCS algorithms to calculate the shortest path, and the results are displayed clearly on the map. It helps users understand route optimization in an easy and visual way.

3. The frontend and backend work smoothly together, and the system responds quickly to user input. The interface is simple and easy to use.

4. Overall, the project is useful and practical. It can be improved in the future by adding more features like real-time traffic and advanced map options.